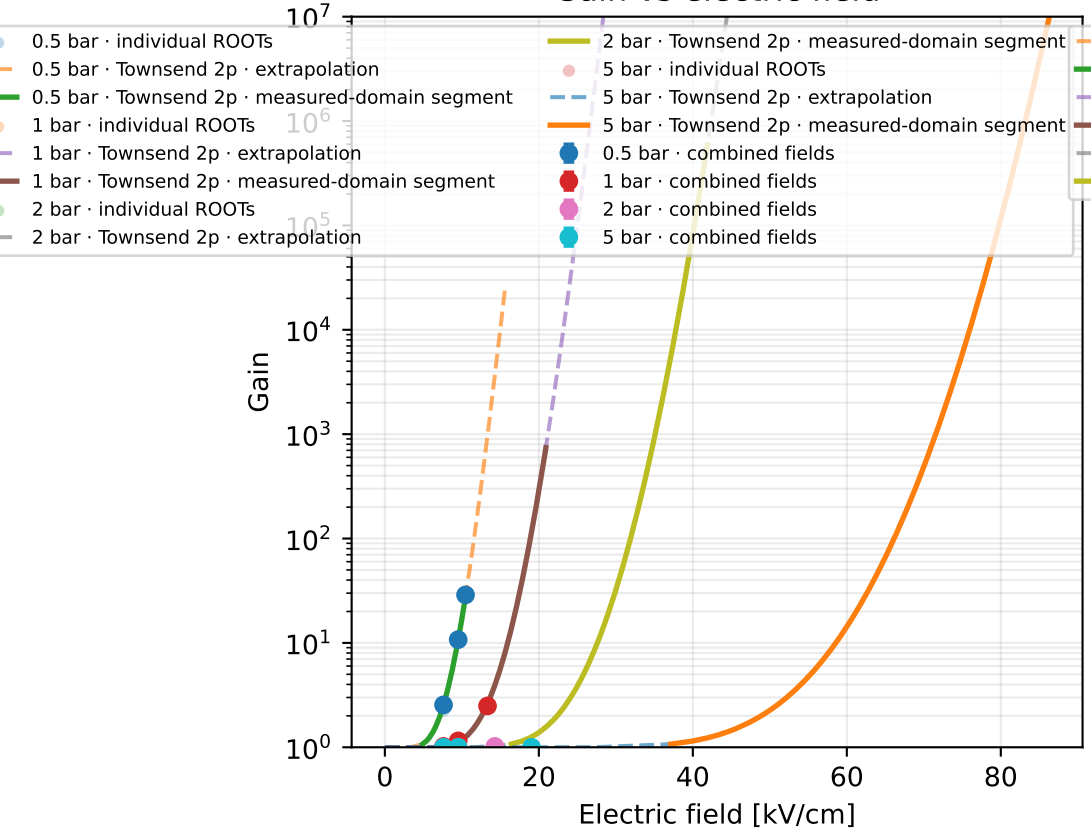
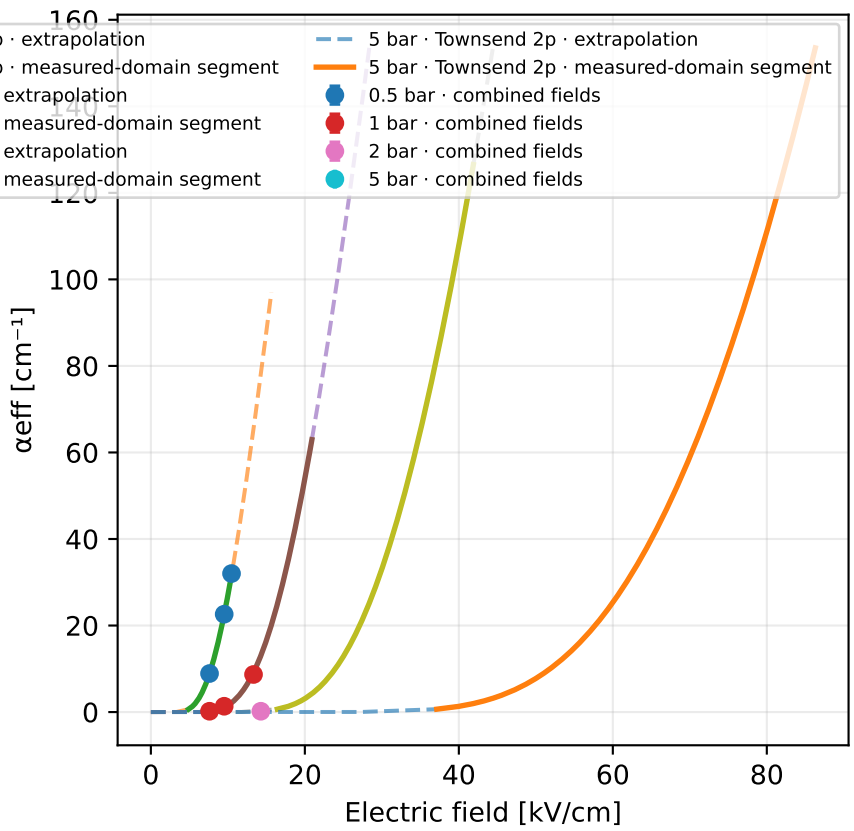
ArN2 · Ar 90 % / N₂ 10 % · gap 1.05 mm

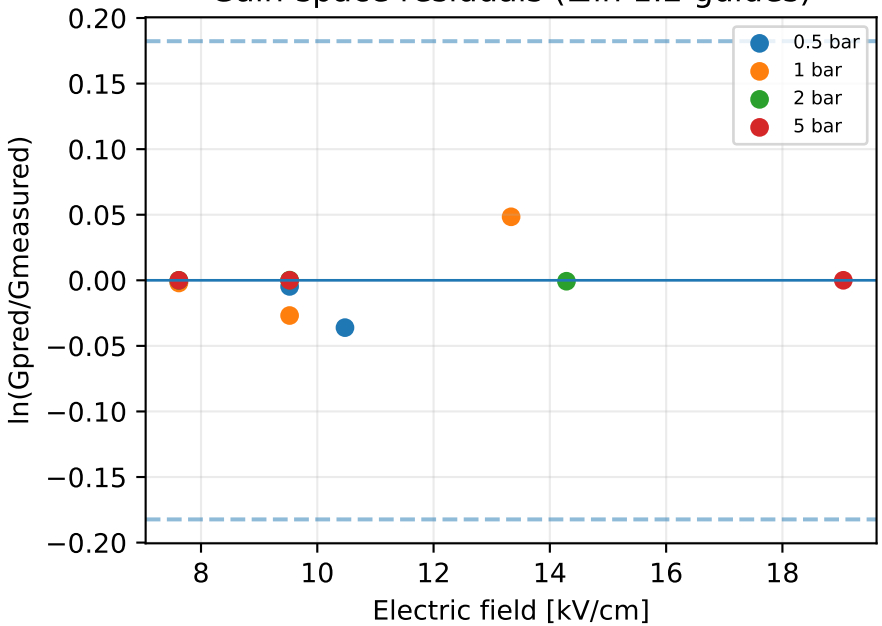
Gain vs electric field



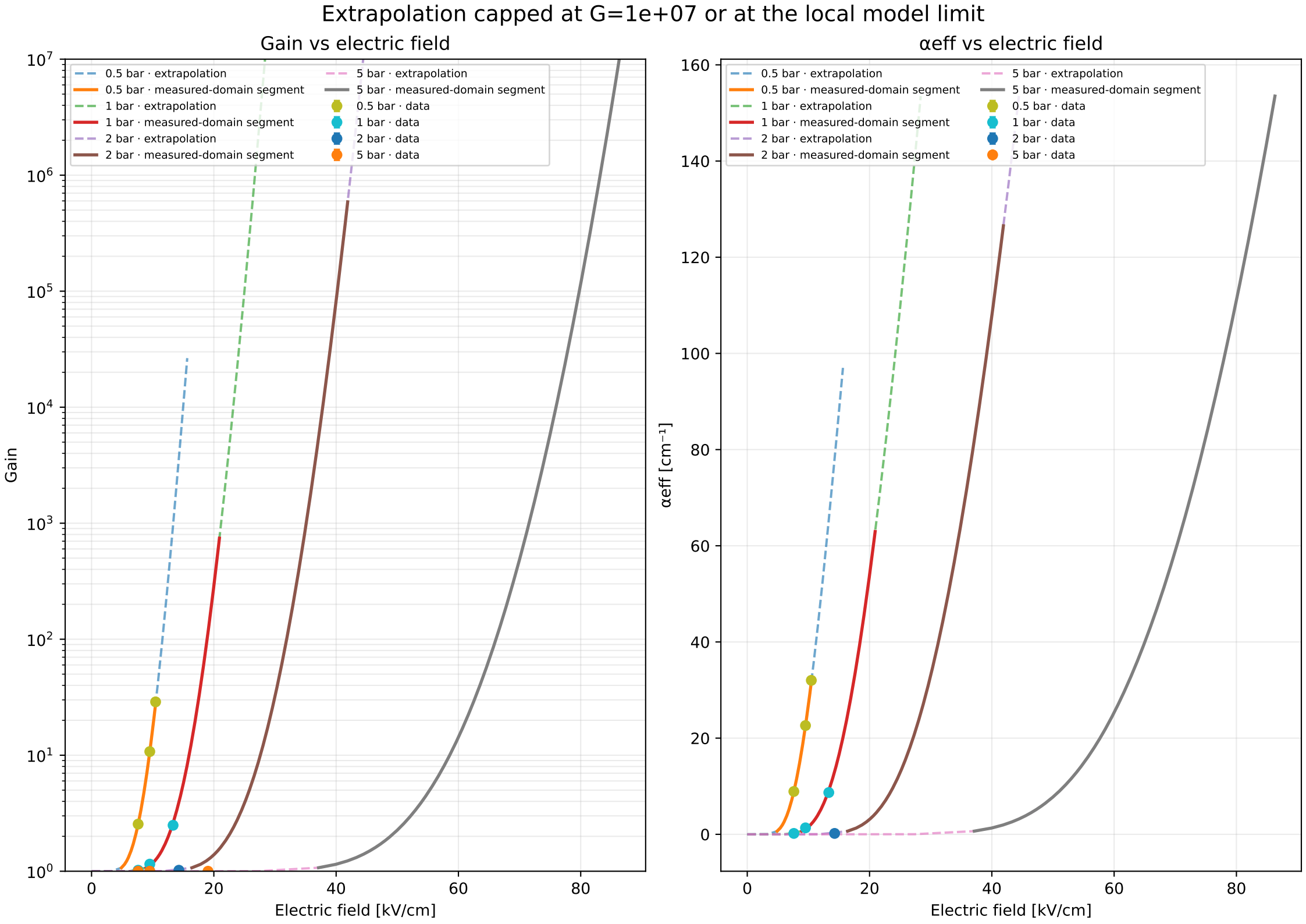
Effective Townsend coefficient vs electric field



Gain-space residuals (±ln 1.2 guides)




```
ROOT runs: 20
Unique physical points: 12
Pressures: 0.5, 1, 2, 5 bar
Selected model: Townsend 2p
Status: accepted
Training RMSE: ×1.097
Cross-validation median: ×1.008
Cross-validation worst: ×1.089
Reduced-field range: 7.143–20.95 kV cm⁻¹ bar⁻¹
A=1871.4, B=70.943, m=0, n=1
Covariance condition: 554
0.5 bar display limit: G=2.65e+04 (local inversion limi
1 bar display limit: G=1e+07 (requested cap)
2 bar display limit: G=1e+07 (requested cap)
5 bar display limit: G=1e+07 (requested cap)
```


Extrapolation capped at G=1e+07 or at the local model limit
Gain vs electric field
αeff vs electric field
0.5 bar · extrapolation
0.5 bar · measured-domain segment
1 bar · extrapolation
1 bar · measured-domain segment
2 bar · extrapolation
2 bar · measured-domain segment
5 bar · extrapolation
5 bar · measured-domain segment
0.5 bar · data
1 bar · data
2 bar · data
5 bar · data
Gain
αeff [cm⁻¹]
Electric field [kV/cm]
Electric field [kV/cm]

Exact combined points used by the fit · page 1/1

Repeated ROOTs at the same field are pooled once; all source ROOTs remain on disk.

p [bar]	E [kV/cm]	E/p	Gain	σGain	αeff [cm⁻¹]	npe	runs
0.5	7.61905	15.2381	2.548333	0.074	8.90895	600	2
0.5	9.52381	19.0476	10.75	0.411	22.6182	600	2
0.5	10.4762	20.9524	28.83167	1.07	32.014	600	2
1	7.61905	7.61905	1.02	0.00572	0.188596	600	2
1	9.52381	9.52381	1.151667	0.0161	1.34486	600	2
1	13.3333	13.3333	2.49	0.0746	8.68841	600	2
2	7.61905	3.80952	1	0	—	500	1
2	9.52381	4.7619	1	0	—	500	1
2	14.2857	7.14286	1.02	0.00619	0.188596	600	2
5	7.61905	1.52381	1	0	—	500	1
5	9.52381	1.90476	1	0	—	500	1
5	19.0476	3.80952	1	0	—	600	2

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-26.3	×1.01	×1.09	yes	—
Townsend 3p	3	-19.8	—	—	no	poor cross validation, large cross validation outlier